

Internship Report

C++ Programming Internship

A hands-on journey through object-oriented programming, file handling, and console application development at Thiranex.

Dinesh Kumar

Intern ID: THX-JUL3126-376 | Role: Intern – C++ Programming

31 July 2026 – 30 August 2026 | Remote / Project-Based



Dinesh Kumar

B.Tech Computer Science
Graphics & Game Technology

ABOUT THE INTERN

Candidate Profile



Intern ID

THX-JUL3126-376



Institute

IILM University, Rajasthan



Internship Role

Intern – C++ Programming



Duration

31 Jul 2026 – 30 Aug 2026 (1 Month)



Work Mode

Remote / Project-Based

Thiranex: Skill Development & Future Tech

Thiranex offered a one-month, remote and project-based internship in C++ Programming. Under industry mentorship, interns work on a series of practical, console-based applications, with progress reviewed periodically to build genuine, job-ready programming skills.



Skill Development

Focused on practical, industry-relevant C++ programming skills.



Industry Mentorship

Regular reviews and guidance throughout the internship period.



Remote & Project-Based

Work delivered independently as a series of graded mini-projects.

PROBLEM STATEMENT

From OOP Theory to Working Applications

Many computer science students learn object-oriented programming, data structures, and file handling in theory, but rarely get to combine them into complete, working applications. Concepts like classes, persistent storage, and menu-driven control flow are often understood on paper without practice turning them into real, testable programs.

This internship closed that gap through four progressively built C++ console applications:



The Toolkit Behind the Internship



C++

Core programming language



OOP Concepts

Classes, objects & encapsulation



File Handling

Persistent data storage



Menu-Driven Design

Structured console navigation



Arrays & Structures

Organizing records in memory



Control Logic

Loops & conditional statements



Testing & Debugging

Validating program behaviour



Console I/O

User input & formatted output

The Four-Task Internship Journey

1

TASK

Due 07 Aug 2026

Student Management System

Console app for student records with file handling

✓ Completed Successfully

2

TASK

Due 14 Aug 2026

Bank Management Application

Simulated core banking operations using OOP

✓ Completed Successfully

3

TASK

Due 21 Aug 2026

Library Management System

Managed books, members & borrowing records

✓ Completed Successfully

4

TASK

Due 28 Aug 2026

Mini Game Project

Tic Tac Toe / Snake — loops, arrays & game logic

✓ Completed Successfully

Featured Projects



Student Management System

A console-based system in C++ to efficiently manage student records using file handling and menu-driven operations.

KEY FEATURES

- Add, update, delete & display student information
- File handling for persistent data storage
- Menu-driven operations for smooth navigation

OUTCOME

A functional application that allows users to add, update, delete, and display student information with persistent data storage for reliable record management.



Bank Management Application

A C++ based application that simulates core banking operations using object-oriented programming and file handling.

KEY FEATURES

- Deposit, withdrawal & balance-check features
- Built using object-oriented programming concepts
- Customer data stored securely via file management

OUTCOME

A secure and functional banking system capable of performing deposits, withdrawals, and balance inquiries while maintaining persistent customer records.

Featured Projects



Library Management System

A system that efficiently manages books, members, and borrowing records using structured, object-oriented programming.

KEY FEATURES

- Structures / classes for book & member details
- Book issue and return functionality
- Search functionality by title or author

OUTCOME

A functional application enabling book addition, issue and return processing, and search by title or author for streamlined library operations.



Mini Game Project (Tic Tac Toe / Snake)

A console-based mini game in C++ that demonstrates core programming concepts such as loops, arrays, and conditional logic.

KEY FEATURES

- Loops, arrays & conditional statements drive gameplay
- Game board displayed dynamically after each move
- Win/loss detection with replay options

OUTCOME

An interactive game with dynamic board display, win/loss detection, and replay functionality, showcasing effective implementation of game logic.

EXPECTED OUTCOMES

Skills & Capabilities Gained

- ✓ Practical command of C++ syntax, control flow, and program structure
- ✓ Applied object-oriented programming: classes, objects & encapsulation
- ✓ File handling for persistent data storage across applications
- ✓ Designing clear, menu-driven console interfaces
- ✓ Working with arrays and structures to organize real records
- ✓ Building, testing, and debugging four complete applications end-to-end

COMPLETION SUMMARY

Internship Successfully Completed



Organization

Thiranex — Skill Development & Future Tech



Role

Intern – C++ Programming



Duration

31 Jul 2026 – 30 Aug 2026



Work Mode

Remote / Project-Based

ALL TASKS COMPLETED



Student Management System

Submitted on track — 07 Aug 2026



Bank Management Application

Submitted on track — 14 Aug 2026



Library Management System

Submitted on track — 21 Aug 2026



Mini Game Project

Submitted on track — 28 Aug 2026

Status: 4 / 4 Projects Completed Successfully

Thank You

C++ Programming Internship | Thiranex

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